Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering
09/2023 – Expected 06/2028

- GPA: 2.93/4.00
- Extracurriculars: Executive Board Member & DJ at Radyo Boğaziçi; Regional Organizational Team Member at ESTIEM; Work and Travel USA participant; Student-Athlete (sailing, skiing & windsurfing), doing wakeboarding, kiteboarding, surfing, bouldering, running, fitness, and traveling as hobbies.
- Volunteer: Volunteering Division Member at Industrial Engineering Club (BÜYAK); provided educational mentorship to high school students.

Kabataş Boys' High School
09/2018 – 06/2023

- GPA: 3.78/4.00
- Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.
- Volunteer: Community Involvement Projects Club and TOÇEV (Tüvana Okuma İstekli Çocuk Eğitim Vakfı).

Experience

Backend Developer Intern at DefineX
09/2026 – Present

- Joining the backend development team to contribute to scalable enterprise software solutions, robust backend architecture, RESTful API design, and system optimizations in a dynamic enterprise environment.

Site Reliability Engineering Intern at Akbank
07/2026 – 09/2026

- Completed a 2-month internship with the SRE team, contributing to the reliability, scalability, and performance of mission-critical banking systems. Engaged in Linux systems engineering, operational monitoring, and high-availability systems management.

Software Engineering Intern at Smash Mate
06/2026 – 07/2026

- Completed an intensive 1-month internship during the critical pre-launch phase, contributing to a complex microservices-oriented backend (Java, Quarkus) and a mobile frontend application (React Native). Developed core backend features, resolved high-priority issues, and authored comprehensive unit/integration tests to ensure zero-defect deployments. Played an active role in agile workflows, code reviews, and system optimizations, leading directly to the product's successful official launch.

Technical Projects

- **AlphaTrigger** — Autonomous HFT simulator executing Z-Score mean reversion with dynamic volatility sizing & scoring.
- **FinTech Platform** — Financial analytics platform featuring a continuous limit-order matching engine & portfolio risk models.
- **Seated** — Multi-tenant SaaS featuring an interactive drag-and-drop floor plan builder, table merging & live queue ops.
- **Performo** — Enterprise administration console with role-based access control, server-rendered analytics & user workflows.
- **Enterprise FAQ** — Full-stack knowledge base with SWR optimistic UI updates, database indexing, role-based auth & API tests.
- **Space Shooter IPC** — Asynchronous 60 FPS IPC engine with non-blocking FIFOs, POSIX signals, dlopen plugins & ring buffers.
- **SchedSim** — Zero-libc CPU scheduling simulator in x86-64 assembly using raw Linux syscalls (FCFS, SJF, SRTF, PF, RR).
- **Solar Escape** — Bare-metal RISC-V RV32I flight guidance on Spike with custom linker memory mapping & greedy pathfinding.
- **CoreMark Profiling** — Benchmarking study evaluating EEMBC CoreMark throughput across compiler flags, ILP & RISC-V extensions.
- **Spireslayer** — Unix command interpreter featuring a custom backtracking recursive descent parser & modular execution state.
- **Aura** — Functional language interpreter supporting first-class closures, currying, AST evaluation & dual scoping engines.
- **MatrixNet** — Graph topology analyzer implementing Tarjan's SCC algorithm & articulation point detection in $O(V + E)$ time.
- **GigMatch-Pro** — Algorithmic marketplace matching platform leveraging custom max-heaps for $O(\log N)$ candidate allocation.
- **Smash Minesweeper** — Mobile game featuring iterative BFS flood-fill, guaranteed safe first-click, chording mechanics & haptics.
- **Qt 2048** — Cross-platform C++ game compiled to WebAssembly featuring directional grid normalization & infinite state undos.
- **Nightpass** — Terminal survival card game driven by a deterministic finite state machine, resource balancing & event queues.

Skills, Tools & Languages

- **Programming Languages:** Python, Java, C, C++, TypeScript, JavaScript, SQL, x86-64 Assembly, RISC-V Assembly
- **Technologies & Frameworks:** Next.js, React Native, Quarkus, FastAPI, Express, Tailwind CSS, Prisma, PostgreSQL, Redis, Docker, WebSockets, WebAssembly, Qt6
- **Tools & Software:** Git/GitHub, Linux/POSIX, AI-Assisted Development, MS Office
- **Languages:** Turkish (Native), English (Advanced), German (Beginner)